

1. What is Ohm's law?
A. $V = IR$ B. $P = VI$
C. $V = I/R$ D. $P = V^2/R$
2. Which material allows the flow of electric current easily?
A. Insulator
B. Conductor
C. Semiconductor
D. Superconductor
3. Which theorem states that the total current in a linear bilateral network is equal to the algebraic sum of the currents in the individual branches?
A. Superposition theorem
B. Thevenin's theorem
C. Norton's theorem
D. Kirchhoff's law
4. According to Thevenin's theorem, what is the equivalent resistance across the output terminals?
A. Zero B. Infinity
C. V/R D. R
5. What is the RMS value of an alternating current or voltage?
A. Root Mean Square
B. Average Value
C. Peak Value
D. Instantaneous Value
6. How many phases are there in a three-phase system?
A. One B. Two
C. Three D. Four
7. What is the basic function of a semiconductor diode?
A. Amplification
B. Rectification
C. Oscillation
D. Modulation
8. Which type of transistor has three layers of semiconductor material?
A. BJT B. FET
C. MOSFET D. JFET
9. What is the function of an oscillator in a circuit?
A. To amplify signals
B. To generate waveforms
C. To modulate signals
D. To rectify signals
10. What is the power (in watts) dissipated by a resistor with a resistance of 50 ohms when a current of 2 amperes flows through it?
A. 25 W B. 50 W
C. 75 W D. 100 W
11. A circuit has two resistors in parallel. One has a resistance of 10 ohms and the other has a resistance of 20 ohms. What is the total resistance of the circuit?
A. 5 ohms B. 6.67 ohms
C. 10 ohms D. 30 ohms
12. In a circuit, the Thevenin equivalent voltage is 24V and the Thevenin equivalent resistance is 6 ohms. What is the current when a load resistance of 4 ohms is connected?
A. 2 A B. 3 A
C. 4 A D. 6 A
13. For a circuit with a maximum power transfer, if the source voltage is 20V and the internal resistance is 5 ohms, what should be the load resistance?
A. 5 ohms B. 10 ohms
C. 15 ohms D. 20 ohms

14. A sinusoidal AC voltage has a peak value of 20V. What is its RMS value?
A. 14.14V B. 15.81V
C. 17.68V D. 20V
15. In a three-phase system, each phase has a voltage of 120V. What is the line voltage?
A. 120V B. 208V
C. 240V D. 360V
16. A diode has a forward voltage drop of 0.7V and a forward current of 10mA. What is the power dissipated in the diode?
A. 0.007 W B. 0.07 W
C. 0.7 W D. 7 W
17. Calculate the base current (I_B) of a BJT transistor with a collector current (I_C) of 5mA and a current gain (beta) of 100.
A. 0.05 mA B. 0.5 mA
C. 5 mA D. 50 mA
18. An RC oscillator has a resistor of 1k ohm and a capacitor of 10nF. What is the frequency of oscillation?
A. 10 Hz B. 100 Hz
C. 1 kHz D. 10 kHz
19. In a Class AB amplifier, the collector current is 20 mA and the collector-emitter voltage is 10V. What is the power dissipated in the collector?
A. 0.2 W B. 0.5 W
C. 1 W D. 2 W